

Explainability Layer

Why the system produced a given output — evidence, confidence, and decision trace

Sentinel: AI Agent Reliability Platform · mechanisms + interface elements

The question every output must answer

For every output, a non-AI engineer can answer four questions without reading code: **(1) what evidence** did the agent use, **(2) how confident** is it and where is it uncertain, **(3) what exact sequence of steps** led here, and **(4) which step is to blame** when something failed — plus the ability to **re-run the exact trajectory** to verify any of it.

Brief §1.5.2: “a mechanism or interface element showing why the system produced a specific output, including evidence, confidence, and decision trace where applicable.” Brief §1.3.4 / judging §5 (15%): outputs are traceable, the system communicates uncertainty, and a non-specialist can understand why and what to do.

1 Five pillars of explainability

Explainability in Sentinel is not a panel bolted on at the end — it is a property of the data structures themselves. Every shared object in `trace_core` carries its own justification, so explanation is produced *by construction*.

Pillar	Mechanism (data structure)	Where you see it
Evidence	<code>SearchResult.attribution</code> (dims, terms, confidence_margin); <code>evidence[]</code> cites ids	AttributionBox, RCA comment in Jira
Confidence	<code>confidence_score</code> , <code>judge_confidence</code> , <code>confidence</code> per-dimension	VerdictCard, ConfidenceBar
Decision trace	ordered <code>TraceRecord[]</code> (embedding → search → search → chat)	StepTimeline, Langfuse
Attribution	<code>FailureAttribution</code> (step, component, description, confidence)	AttributionBox, recommended action
Audit & replay	hash-chained <code>AuditBlock</code> ; deterministic <code>replay_link</code>	replay screen, audit chain in D1

2 Anatomy of an output — the EvalVerdict

The product of evaluation is a single structured, human-readable `EvalVerdict`. Every field exists to be *acted on*, not just read:

```
EvalVerdict -- run AO-11 (trial 0)
```

```
verdict: uncertain trial_number: 0
dimensions: correctness 0.90 · efficiency 0.64 · safety 0.92 · reasoning_quality 0.78
failure_attribution: { step 1, component retrieval, "two judge passes disagreed on this case", confidence 0.70 }
self_evaluation: { judge_confidence 0.66, self_critique "rubric order changed the verdict; needs review", flag_for_human true }
recommended_action: "Judge verdict is order-sensitive (position bias); human review required."
replay_link: dashboard.ahmedxsaad.me/replay/<run_id>
```

Read it in one breath: the run is *uncertain*, the weak dimension is *efficiency* (0.64), the judge is not confident (0.66) so it was *flagged for a human*, and the recommended action tells the engineer exactly what to do next — with a link to re-run it. No AI expertise required.

3 Pillar 1 — Evidence (what it used)

Retrieval is never a black box. `xqdrant` returns, for every hit, an `Attribution` block: per-dimension and per-term contributions to the cosine score, and a `confidence_margin` — the gap to the next-best hit. A small margin means an ambiguous match, which is itself a signal surfaced to the user.

```
SearchResult (incidents) -- top hit for AO-11
```

```
id: AO-11-src score: 0.99
text: "Connection pool exhaustion cascading to all services"
attribution.confidence_margin: 0.27 (strong, unambiguous match)
attribution.terms: { "connection pool": 0.41, "cascading": 0.22, "exhaustion": 0.19 }
```

The drafted `RcaDraft` then *cites* this evidence by id in its `evidence[]` list and records `knowledge_gaps[]` when no runbook clears the relevance floor — so the final Jira comment shows not just a conclusion but the grounding behind it.

4 Pillar 2 — Confidence and uncertainty

The system communicates uncertainty at three nested levels, and *degrades gracefully* rather than acting when unsure:

- **Generation** — `RcaDraft.confidence_score`; below `CONFIDENCE_THRESHOLD = 0.70` the run is flagged for a human instead of auto-acting. Duplicate auto-linking uses a stricter 0.85 gate.
- **Per-dimension** — each `DimensionScore` carries its own `confidence` alongside its score and a free-text `reason`.
- **Verdict** — `SelfEvaluation.judge_confidence` plus a `self_critique`. When the calibrated judge is order-sensitive, the verdict becomes *uncertain* (not a coin-flip) and `flag_for_human` is set.

Self-evaluation (bonus, brief \$5). The judge evaluates its own verdict and surfaces low-confidence results to a human reviewer — the system knows, and says, when it does not know.

5 Pillar 3 — Decision trace (the exact sequence)

Every run is a non-lossy, ordered list of `TraceRecord` steps captured at the network boundary by the UC2 flight recorder — nothing is reconstructed after the fact. For AO-11:

#	Step	Model / tool	Readable preview
0	embedding	bge-base-en-v1.5	“embed 1 text ...” → 768-dim vector
1	vector search	xqdrant incidents	top hit 0.99 (connection-pool exhaustion)
2	vector search	xqdrant runbooks	“DB connection pool exhaustion” 0.72
3	chat (RCA)	llama-3.1-8b	hypothesis + cited evidence + severity

Each step stores a human-readable input/output preview (not raw JSON), the model id, and the operation, so the dashboard `StepTimeline` reads as a story. The same calls are exported to **Langfuse** as GenAI traces for an engineer who wants the raw prompt/response/latency.

6 Pillar 4 — Failure attribution (who is to blame)

A failing run is mapped onto a small DAG, `retrieval` → `planning` → `execution`, and the first step showing a failure signal is blamed — with a confidence graded by how explicit the signal is:

Signal	Confidence	Component
explicit error / <code>success=false</code>	0.90	as classified
retrieval returned no results	0.70	retrieval
execution produced no key/id outcome	0.70	execution
no explicit signal (blame last step)	0.50	fallback

The output is a sentence, not a stack trace: “*Step 2, retrieval, returned no results (confidence 0.7)*” — the difference between “it failed” and “open replay and fix the retrieval step.”

7 Pillar 5 — Audit and deterministic replay

Explainability is only trustworthy if the record cannot be quietly altered and the claim can be reproduced. Two mechanisms guarantee this:

- **Tamper-evident audit** — every step is an `AuditBlock` with a `payload_hash` (SHA-256 of canonical JSON), an `hmac` (signed with a server key), and a `prev_hash` linking it to the previous step. Written *write-ahead* (before execution), the chain proves the trajectory was not edited after the fact.
- **Deterministic replay** — `replay_link` re-runs the exact trajectory from the cassette with **zero live calls**; a request the agent never originally made shows up as a *divergence*. `bisect` finds the first diverging step between two runs, and `mid-replay inject` lets an engineer change one response and watch the path change — explaining *counterfactually* why a step mattered.

8 Interface elements — where to look

The mechanisms above surface in the dashboard (`dashboard.ahmedxsaad.me`). A reviewer never needs the code:

Question	Screen	Interface element
What did the agent do?	/runs/<id>	StepTimeline (operation-aware)
Did it do it right?	/verdicts/<id>	VerdictCard + DimensionTable
Why this verdict?	/verdicts/<id>	AttributionBox + self-critique
Can you prove it?	/replay/<id>	replay (0 live calls) + bisect + inject
How reliable overall?	/ & /reliability	ReliabilityRing, drift, κ
Raw LLM calls?	Langfuse	per-call prompt / response / latency

9 Worked example — AO-11 end to end

1. Evidence — incident embedded once; xqdrant returns “connection-pool exhaustion” at 0.99 (margin 0.27) and a DB-pool runbook at 0.72. **2. Decision trace** — four taped steps (embed → search → search → RCA), each replayable. **3. Confidence** — the RCA cites those ids; the two calibrated judge passes disagree, so judge_confidence drops and the verdict is uncertain. **4. Attribution** — the unstable step + component are named; **5. Audit/replay** — the run replays with 0 live calls and an A0 Jira Incident is auto-filed for the human. Every claim above is a field a reviewer can open and check.

The question every output must answer

Bottom line. Explanation is structural, not cosmetic: evidence with attribution, three levels of confidence, a non-lossy decision trace, per-step blame, and a tamper-evident, re-runnable record — each surfaced as a concrete interface element a non-specialist can act on.